
CAPSTONE PROJECT

AI-POWERED STUDY BUDDY

Presented By:

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OUTLINE

- **Problem Statement** (Should not include solution)
- **Proposed System/Solution**
- **System Development Approach** (Technology Used)
- **Algorithm & Deployment**
- **Result**
- **Conclusion**
- **Future Scope**
- **References**

PROBLEM STATEMENT

- **Information Overload:** Students are regularly overwhelmed by massive textbooks, lengthy research PDFs, and extensive study materials.
- **Lack of Personalization:** Traditional learning methods follow a one-size-fits-all approach, making it hard for individuals to test themselves or get immediate, customized conceptual clarity.
- **Time Inefficiency:** Manually creating revision tools—such as flashcards, core concept summaries, comprehensive quizzes, and organized weekly schedules—takes up valuable hours that could be spent studying.

PROPOSED SOLUTION

- **The AI-Powered Study Buddy:** A unified, interactive educational platform designed to streamline student workflows, boost retention, and act as a 24/7 personal tutor.
- **Core Functionalities:**
 1. **Explain Topics:** Instantly breaks down complex, multi-layered concepts into simple, digestible explanations.
 2. **PDF Notes Summarizer:** Extracts core knowledge and condenses massive documents seamlessly.
 3. **Quiz & Flashcard Generator:** Automated active-recall tools to instantly evaluate topic comprehension.
 4. **Study Plan Generator:** Instantly maps out organized structured schedules for dynamic preparation.

SYSTEM APPROACH

- **Core Language: Python** — Utilized for processing logic, managing backend operations, and integrating core API frameworks.
- **Frontend UI Framework: Streamlit** — Powering a responsive, clean, and highly interactive user interface designed for immediate data updates.
- **Generative AI Model: Google's Gemini AI** — Acts as the central large language model engine handling contextual reasoning, smart summarization, and query generation.
- **Document Parsing: PyPDF2** — Integrated to read, parse, and handle structured textual data directly from uploaded PDF files.

ALGORITHM & DEPLOYMENT

Backend Workflow Pipeline

[User Input: Topic / PDF] → [Streamlit UI Processing] → [PyPDF2 Text Extraction (if PDF)] | [Download Output Data]  [Render Streamlit Output] ← [Gemini AI Processing Engine]

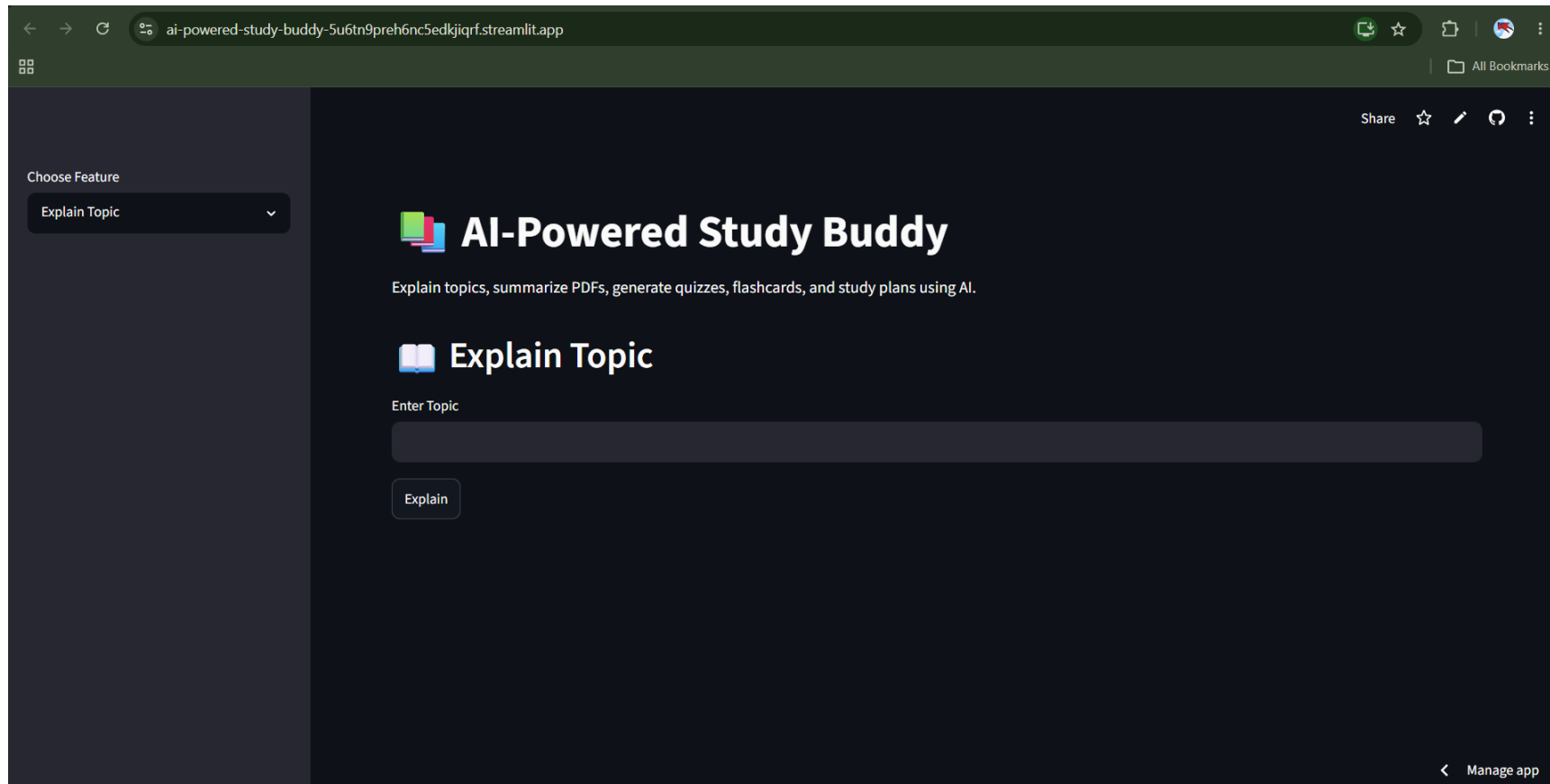
Deployment Architecture

1. Hosting Platform: Deployed directly onto **Streamlit Community Cloud** to offer an instantly accessible, serverless web app layout.

2. Execution Flow: The user triggers an event via the sidebar dropdown navigation interface → Backend routes the specific prompt structure directly to the Gemini API → Results are generated, formatted into clean markdown blocks, and cached seamlessly for immediate download.

RESULT

- **Functional Deliverable:** A fully operational, high-performance web interface allowing smooth navigation between separate modular features via an integrated sidebar dashboard.
- **User Interface Showcase:**



CONCLUSION

- **Successful Integration:** Effectively coupled Large Language Models (LLMs) with intuitive web design to solve real-world student productivity gaps.
- **Enhanced Productivity:** Drastically cuts down on textbook scanning and manual study-aid creation time, pivoting the student focus onto high-impact active learning.
- **Seamless Accessibility:** The web application architecture ensures that students can access tailored educational support on-demand from any connected device.

FUTURE SCOPE

- **User Authentication & Cloud Storage:** Implementing individual user profiles to securely store past study history, generated flashcards, and uploaded PDFs..
- **Multi-Format Support:** Expanding text processing capabilities to read handwritten notes, audio lectures, and image-based charts or diagrams.
- **Advanced Analytics:** Adding a performance dashboard tracking quiz accuracy trends to pinpoint a student's weak areas over time.

REFERENCES

Project Repository: [<https://github.com/adarshprajapati007/AI-Powered-Study-Buddy>].

Live Web Application: <https://ai-powered-study-buddy-5u6tn9preh6nc5edkjiqrf.streamlit.app>

Documentation & Libraries:

- Python Official Documentation (python.org)
- Streamlit Framework Framework (streamlit.io)
- Google Gemini AI Platform Documentation

GITHUB LINK

- Github Link : <https://github.com/adarshprajapati007/AI-Powered-Study-Buddy>
- Deployment Link: <https://ai-powered-study-buddy-5u6tn9preh6nc5edkjiqrf.streamlit.app/>



THANK YOU